Live Transcription with Review

Making voice input feel as controlled and precise as typing

Team: Product / Design / Engineering
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Status: In Review
Launching on: Week 13–14 (post A/B test validation)
Resources: Figma Wireframes / Mind Map / User Research Deck / UX Audit & Competitive Benchmarks
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Problem Statement: Mobile-first Indian professionals avoid voice input because the capture phase operates as a black box — providing no transparency, no ability to catch errors in real time, and no control before submission.
Target Segment: Mobile-first Early-Career Professionals (18–35) in India using ChatGPT for high-stakes tasks — technical upskilling, coding, professional drafting, and interview prep.
Success Metric: 10–15% **relative** increase in voice input completion rate within 3 months — baseline to be confirmed with Data & Analytics pre-launch.

Problem Definition:

Mobile-first Indian professionals (18–35) perceive ChatGPT's voice input as a risky "one-shot" interaction because it lacks the visual verification, editing control, and drafting confidence of typing. This problem is especially significant in India's oral-first mobile ecosystem, where long-form typing is slow and mentally tiring. Research with 47 users revealed that 36% avoid voice input due to a lack of trust in what is being captured before submission. By creating a voice experience that feels editable and controllable like a keyboard, users can capture complex thoughts faster with lower cognitive effort and fewer errors. Solving this now can drive higher voice adoption, deeper engagement, and long-term retention in one of the world's largest mobile-first markets.

Goals :

- **Increase voice input adoption** — More users should choose voice over typing for high-stakes tasks within the target segment.
- **Build user trust in voice input** — Users should feel confident that what they speak will be captured and processed accurately before submission.
- **Reduce friction in the voice input experience** — The effort required to use voice should feel comparable to or lower than typing.
- **Establish voice as a habitual input method** — Users should return to voice repeatedly, not just try it once.
- **Maintain product quality during rollout** — The existing ChatGPT mobile experience should not degrade as the new feature ships.

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Functional Metrics	
Metric	**Why it matters**
Voice input adoption rate	Primary success metric — are more users trying dictation after the feature ships
CTR on the mic button after seeing the live transcript option	Measures whether visibility of the feature alone reduces upfront hesitation to try voice
Voice input completion rate	Measures how many users who tap mic actually follow through and send — drop-offs indicate trust still broken
Edit rate post-transcription	How often users correct the transcript before sending — high edit rate validates the feature is being used as intended
Repeat voice usage rate(Retention)	Are users coming back to voice after first use — signals trust is being built over time
Live Transcript toggle OFF rate	OFF rateHow many users disable it — high toggle-off rate signals the feature feels intrusive rather than helpful

Non-functional Metrics	
Metric	**Why it matters**
Transcription latency	If words don't appear fast enough to feel live, the core trust mechanism breaks
Speech-to-text error rate	High error frequency exposes the problem rather than solving it
App crash rate during voice session	A crash mid-dictation is the worst possible trust-breaking moment

Non-Goals:

- Voice perceived as slower — separate solution track
- Advanced Voice Mode users — outside scope
- Social & Environmental Friction — 26% avoid voice in public/office, 19% cite social awkwardness — UI change cannot fix physical/social context
- Accent & Jargon Misinterpretation — 17% affected — live transcription makes errors visible but doesn't reduce their frequency. Requires model-level fix
- Noisy environment accuracy — hardware/model limitation, outside scope

Validation of the Problem

User Research (n=47)

- Blind-Input Trust Gap is the single largest barrier to voice adoption — affecting 36% of users, ahead of noise, privacy, and accent concerns
- Users don't distrust voice — they distrust invisibility during capture
- Two user quotes make the root cause concrete:
  - *"I need to think while I type"* — speaking feels like a final performance with no safety net
  - *"One mistake ruins the prompt"* — one mishearing of a technical word is enough to abandon voice entirely and default back to typing.

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Competitive Research

- Live transcription during capture already exists across tools users interact with daily — iOS Native Dictation, Google Docs, Otter.ai
- ChatGPT is the outlier for not surfacing it — the solution is proven elsewhere, the gap just hasn't been closed here yet

Understanding the target audience:

Segment in Focus:
Mobile-first Early-Career Professionals (18–35) in India using ChatGPT as a high-precision career co-pilot for technical upskilling, professional drafting, coding, and mock interview prep. This segment scored highest across size, potential to change, and accessibility - making it the highest-leverage starting point for voice adoption.

Size:
Represents the core of India's digital workforce — the demographic driving the highest daily active usage on ChatGPT mobile. Operating in an oral-first economy of 950M+ mobile users where long-form typing on small screens causes real physical and cognitive fatigue.

Key Personas:
Three distinct user types within this segment:

The Upskiller — refining code, verifying technical logic, debugging on the move. Needs precision. One misheard technical term breaks the entire prompt.

The Career Climber — prepping for interviews, practising mock responses, crafting referral messages. High stakes. Every prompt matters.

The Professional Drafter — writing portfolio summaries, stakeholder communications, complex emails. Needs editability and control before anything is sent.

Unmet Needs
Goals:

- Speak complex technical thoughts faster than they can type them
- Use voice for high-stakes tasks without fear of losing precision
- Feel as in control of voice input as they do of a keyboard

Pain Points:

- Voice feels like a one-shot performance with no safety net
- No visibility during capture — can't catch errors in real time
- One mishearing of a technical word forces them back to typing entirely

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Solution:

Solution Directions:

Direction 1 — Live Transcription with Review

The idea: Replace the black box with a visible, editable transcript that the user controls before sending.

Steps:

User taps waveform->Words appear in the text field in real time as the user speaks — solid text, no ambiguity->User stops speaking -> Full transcription is now sitting in the text field, exactly like typed text -> User scans it, edits anything that looks wrong -> User taps send

What this solves: Trust gap — user sees everything, controls the send moment, correction costs nothing.

[Make live transcription permanent and visible by default, with an option to minimise it for those who prefer less visual noise]

Direction 2 - Audio playback confirmation

The Idea: Let users hear what was captured before sending, so trust is built through voice confirmation rather than text confirmation.

Steps:

User taps mic -> User speaks naturally -> User taps waveform to stop ->The app plays back the recorded audio -> User listens and checks whether it matches their intent -> User edits if needed, then taps send

What it solves: This direction solves trust through hearing instead of reading. It is useful for users who care more about tone, phrasing, or whether the app actually caught what they said, rather than reviewing text line by line.

Direction 3 — Loose Speech Processing with Post-Capture Review

The idea: Users don't just distrust whether the system heard them correctly — they distrust whether it understood what they *meant*. This direction closes that second layer of the trust gap by keeping the capture phase simple and distraction-free, then making the system's interpretation fully visible and verifiable before anything is sent.

Steps:

User taps mic -> Waveform appears — a contextual nudge appears beneath it: *"Just talk naturally, we'll handle the rest."* -> User speaks freely, no pressure to be structured or articulate

No live transcription — user just speaks, waveform confirms the mic is actively picking up voice

User taps waveform to stop ->System presents two versions:

Raw transcript — exactly what was spoken, word for word

Interpreted prompt — the cleaned, structured version the system will actually process

User compares both — if the interpreted version misses the intent, they can edit it directly -> User taps send

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What this solves: The capture phase stays clean and focused — no text appearing mid-speech to distract or overwhelm. Trust is built *after* capture, not during it. The user sees exactly what was heard and exactly what will be processed — two separate confirmations before anything is sent. Control comes from the deliberate stop and the explicit review moment, not from monitoring in real time.

Why these are mutually exclusive:
These three directions are different because each one relies on a different trust mechanism: Seeing it live. Controlling it at the end. Hearing it back.

Solution Directions	Impact	Effort	Confidence
Live Transcription with Review	High	Medium	High
Audio playback confirmation	Medium	Medium	Medium
Loose Speech Processing with Post-Capture Review	Medium	High	Medium

Chosen Solution Direction Overview:

Live Transcription with Review — making voice feel as controlled and precise as typing.

How this draws from research

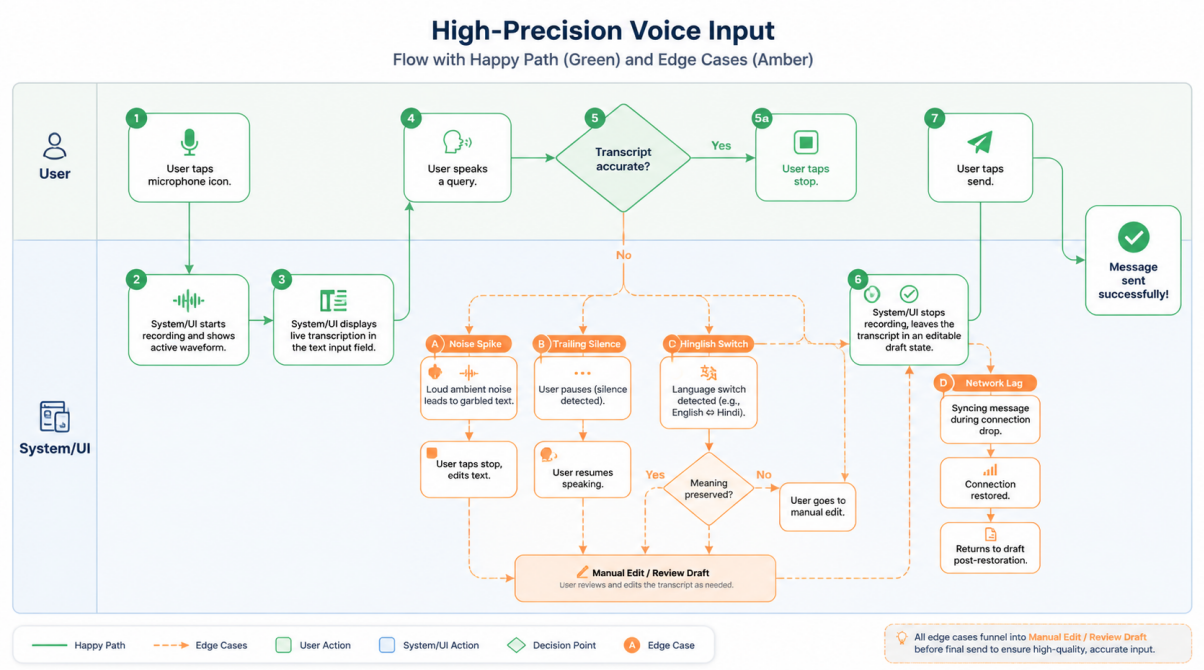
User Research revealed that the trust gap is psychological, not technical — users don't distrust voice, they distrust invisibility. The two mental models — *"I need to think while I type"* and *"one mistake ruins the prompt"* — directly shaped the solution's core mechanic: make the capture phase visible and make correction feel cheap.

Competitive Research showed that live transcription during capture already exists natively on every iPhone (iOS dictation), Google Docs, and Otter.ai. ChatGPT is the outlier for not surfacing it. The solution is proven elsewhere — the gap just hasn't been closed here yet.

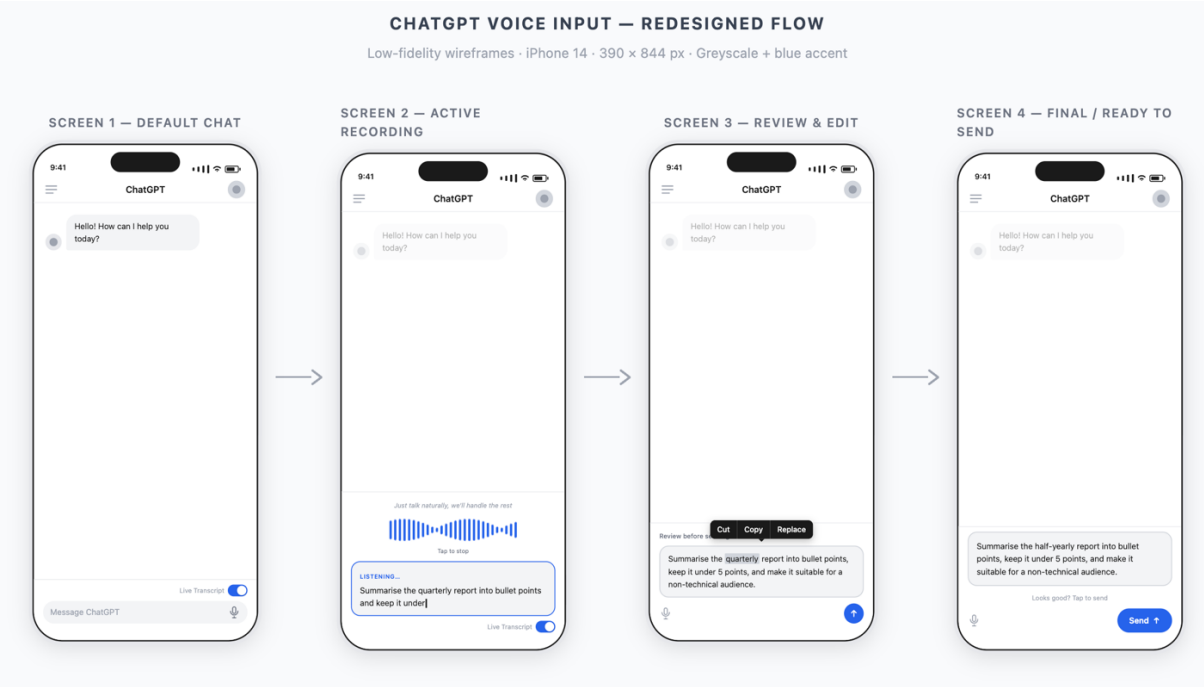
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User Flow

Tap mic → waveform appears + nudge *"Just talk naturally"* → words appear live in text field → tap waveform to stop → review transcript → edit if needed → tap send



(Wireframe: 4 screens — Default, Active Recording, Review & Edit, Ready to Send)



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Key Features

Key Features — Live Transcription with Review

Feature	User Benefit
Live Transcription	Words visible in real time — removes black box anxiety during capture
Contextual Nudge	"Just talk naturally" — reduces composition pressure before speaking
Deliberate Send	No auto-send — user owns the submission moment completely
Live Transcript Toggle	ON by default, user can minimise — permanent, never removed after trust is built
Inline Edit Before Send	Edit directly in the text field — correction feels as cheap as hitting backspace

Key Logic

Key Logic

Algorithm Changes:

- Real-time speech-to-text processing — streaming, not batch. Words must appear as spoken, not after
- Silence detection triggers review state, not auto-send
- Session state transitions — tap waveform triggers recording → review, tap send triggers review → sent
- Nudge trigger logic — first use and hesitation detection only, tracks cross button dismissals, suppressed for confident users
- Language detection — show nudge in user's interaction language, fallback to English if unsupported
- Fallback if STT fails or latency exceeds 200ms — show partial transcript captured so far, display loading indicator
- Confidence scoring per word — foundation for future ghost text implementation

Schema Changes:

- Store raw transcript separately from final sent prompt — enables edit tracking
- Log edit events (word changed, before/after) — feeds future STT accuracy improvement
- Toggle state per session — was Live Transcript ON or OFF
- Timestamps — mic tap to send, measures actual voice session duration

New Data Types:

- Voice session state: recording / review / sent
- Edit delta: tracks corrections between raw transcript and final send
- Nudge trigger state: first use / hesitation detected / suppressed

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- Button seen vs button clicked — two separate events for accurate CTR calculation
- Popup dismissed — tracks cross button clicks separately from nudge being ignored
- Time to send — timestamp delta between mic tap and send tap

Launch Readiness

Key Milestones — Live Transcription with Review

Phase	Milestone	Timeline
Discovery	Problem validated, segment confirmed, solution directions finalised	**Week 1–2**
Design	Wireframes complete, user flow annotated, design review signed off	**Week 3–4**
Development	Live transcription streaming integrated, toggle built, inline edit enabled, nudge logic implemented	**Week 5–8**
QA	Functional testing across devices (iOS/Android), accent and jargon stress testing, latency benchmarking	**Week 9–10**
Dogfooding	Internal rollout to ChatGPT team — collect edge cases, fix critical bugs	**Week 11**
Soft Launch	5–10% user rollout, monitor metrics, collect feedback	**Week 12–13**
Full Launch	100% rollout if metrics hold	**Week 14–16**

Launch Checklist

Engineering

- Streaming STT latency under 200ms — words must feel live, not delayed
- Toggle state persists across sessions — user preference is remembered
- Edit delta logging is live and capturing correctly
- Voice session state transitions are stable across all 4 screens
- Tested on low-bandwidth networks — critical for Tier 2/3 users

Design

- Dark mode UI matches existing ChatGPT mobile design language
- Nudge copy finalised and localisation considered
- Toggle placement consistent across all screens
- Waveform button unchanged — existing UI untouched

Legal & Privacy

- Raw transcript storage compliant with data privacy policies
- Edit delta data anonymised before logging
- User informed that transcription is processed in real time

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Support & Operations

- Support team briefed on new feature — common user questions anticipated
- In-app feedback mechanism live for voice sessions
- Rollback plan defined if latency issues surface at scale

Internal Stakeholders

Internal Stakeholders — Live Transcription with Review

Stakeholder	Interest
Engineering	Streaming STT integration, latency, stability
Design	UI consistency, nudge copy, toggle behaviour
Data/Analytics	Edit delta tracking, metric dashboards live before launch
Legal/Privacy	Transcript storage, data anonymisation
Support	User-facing FAQ, issue escalation path
Marketing	Feature communication — what's new and why it matters
Trust & Safety	Ensuring voice data handling meets safety standards

Experimentation Plan

Hypothesis: Users who see live transcription during voice capture will have a higher completion rate and higher repeat voice usage rate than users who do not.

A/B Test Design — Live Transcription with Review

	Control Group	Test Group
Experience	Current ChatGPT voice — transcription appears after response	Live transcription during capture + review before send
Split	50% of users	50% of users
Duration	4 weeks	
Size	Minimum 1,000 voice sessions per group for statistical	significance

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Primary Metrics to Watch:

- Voice input completion rate
- CTR on mic button
- Repeat voice usage rate within 7 days

Secondary Metrics:

- Edit rate post-transcription
- Time to send via voice
- Toggle OFF rate

Decision Rule:

- Ship if completion rate and repeat usage rate show statistically significant improvement
- Iterate if toggle OFF rate is high — signals live transcription feels intrusive
- Kill if latency metrics degrade or crash rate increase.

Success Metrics:

Primary Metrics:

- Voice input completion rate — 10–15% relative increase within 3 months
- Weekly voice usage rate — 10% relative increase within 3 months
- CTR on mic button after seeing Live Transcript toggle — baseline to be confirmed, directional increase expected

Secondary Metrics:

- Edit rate post-transcription — target >20% of voice sessions show at least one edit, confirming the review step is being used
- Live Transcript toggle OFF rate — target below 15%, anything higher signals the feature feels intrusive

Open Questions & Decisions Taken:

Open Questions

- Should the nudge "Just talk naturally, we'll handle the rest" be localised based on the user's previous interaction language — and which Indian languages are prioritised first?
- Should the Live Transcript toggle be ON by default or opt-in — faster adoption vs slower but trust-preserving rollout?
- At what silence duration after mic tap do we classify a user as hesitant and trigger the nudge — and does this threshold vary by user history?
- Ghost text with confidence highlighting was descoped in favour of solid live transcription — should low-confidence word flagging be reconsidered as a Phase 2 A/B test?

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Trade-offs

- Default ON over opt-in — faster adoption but users encounter the feature before choosing it. Opt-in is safer but slows validation.
- Deliberate send over auto-send — full user control preserved but adds one extra tap per session.
- Dictation mode only in v1 — keeps scope clean and experiment isolated but conversation mode users remain unserved.
- UI transparency over STT retraining — live transcription makes errors catchable but doesn't reduce their frequency. Accuracy itself does not improve in v1.

Descoped

- Phase 2: Ghost text with confidence highlighting, yellow flagging for uncertain words, habit nudge after repeateful sessions
- Phase 3+: Hindi/Hinglish live transcription, STT model retraining via edit delta data, Advanced Voice Mode integration

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